

ADVANCED DATA MINING: AMAZON FASHION REVIEWS

From Unstructured Text to Actionable Insights

GROUP 1

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The Challenge: Scaling Human Insight

The Core Problem

Online retailers rely on feedback, but **volume** renders manual analysis impossible.

Total Reviews: **170,000+**
Human Capacity: **Limited**



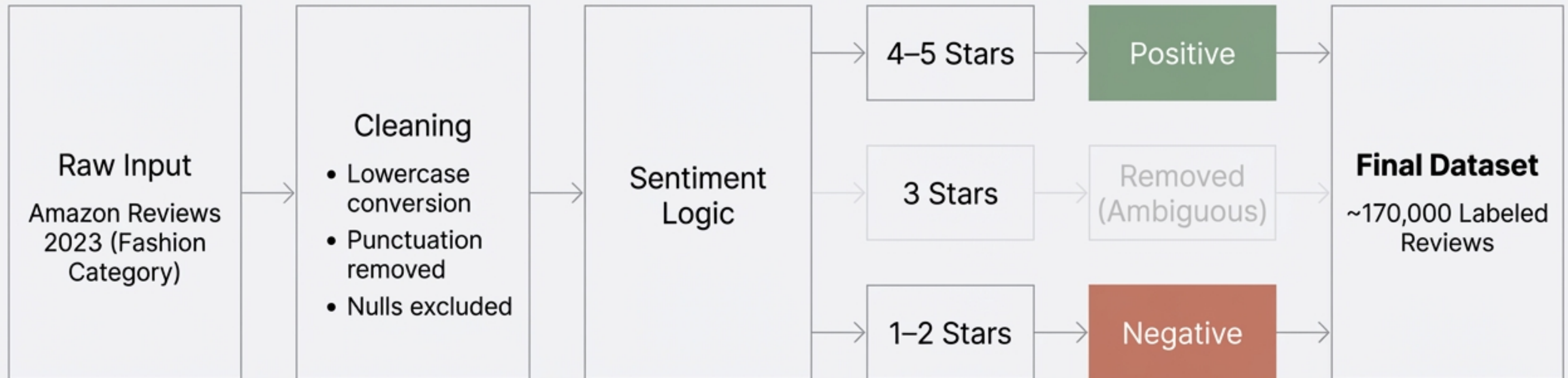
The Objective

Apply text mining to automate the extraction of structure from unstructured text.

Key Research Goals

- 1. Measure review similarity.
- 2. Group reviews into meaningful clusters.
- 3. Classify sentiment using machine learning.

Data Preparation and Logic



Phase I Methodology: Measuring Similarity

Goal: Identify reviews that 'sound' the same to detect redundancy.

Technique

TF-IDF Vectorization

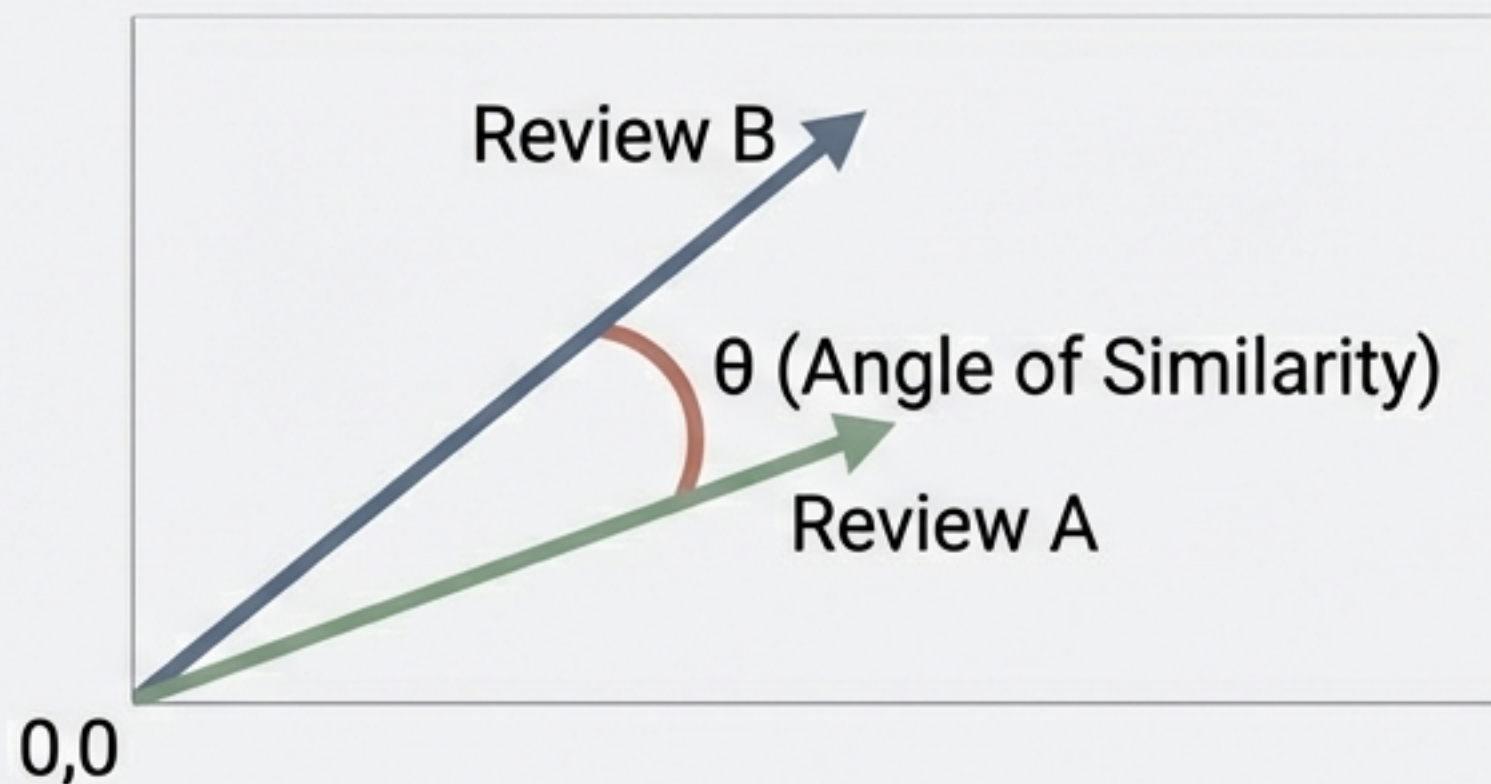
(Term Frequency-Inverse Document Frequency)

Assigns statistical weight to words that are frequent in a specific review but rare across the entire dataset.

Measurement

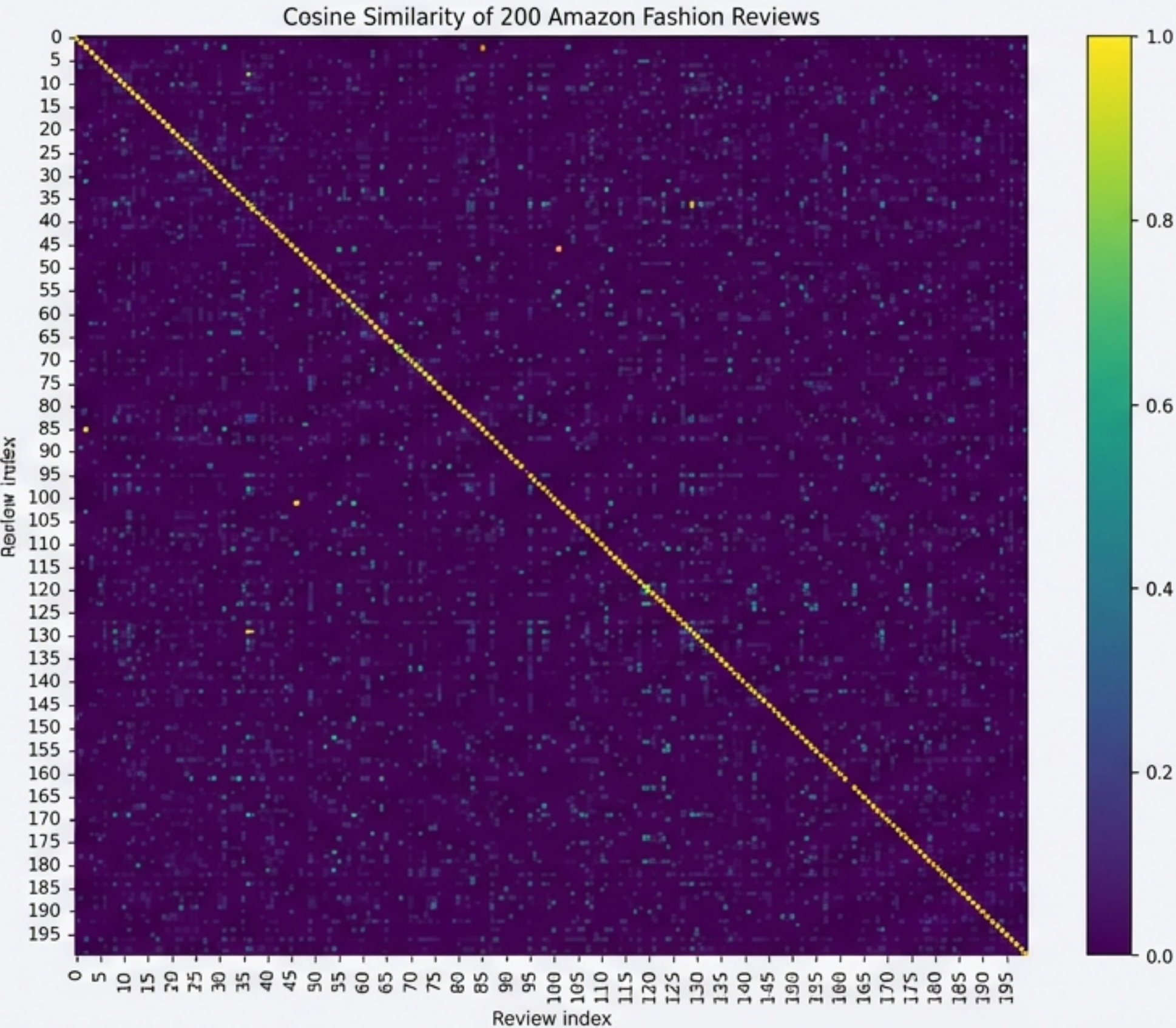
Cosine Similarity

Calculates the cosine of the angle between two review vectors to determine closeness.



Scope: Analysis of 200 representative reviews pairwise.

Visualizing Review Relationships



The Diagonal

Represents perfect self-similarity (1.0).

Hot Spots (Off-Diagonal)

Reveal clusters of reviews sharing significant vocabulary.

Identified Themes

High-similarity pairs frequently discussed:

- Product Fit
- Comfort
- Color
- Overall Satisfaction

Phase II Methodology: Unsupervised Clustering

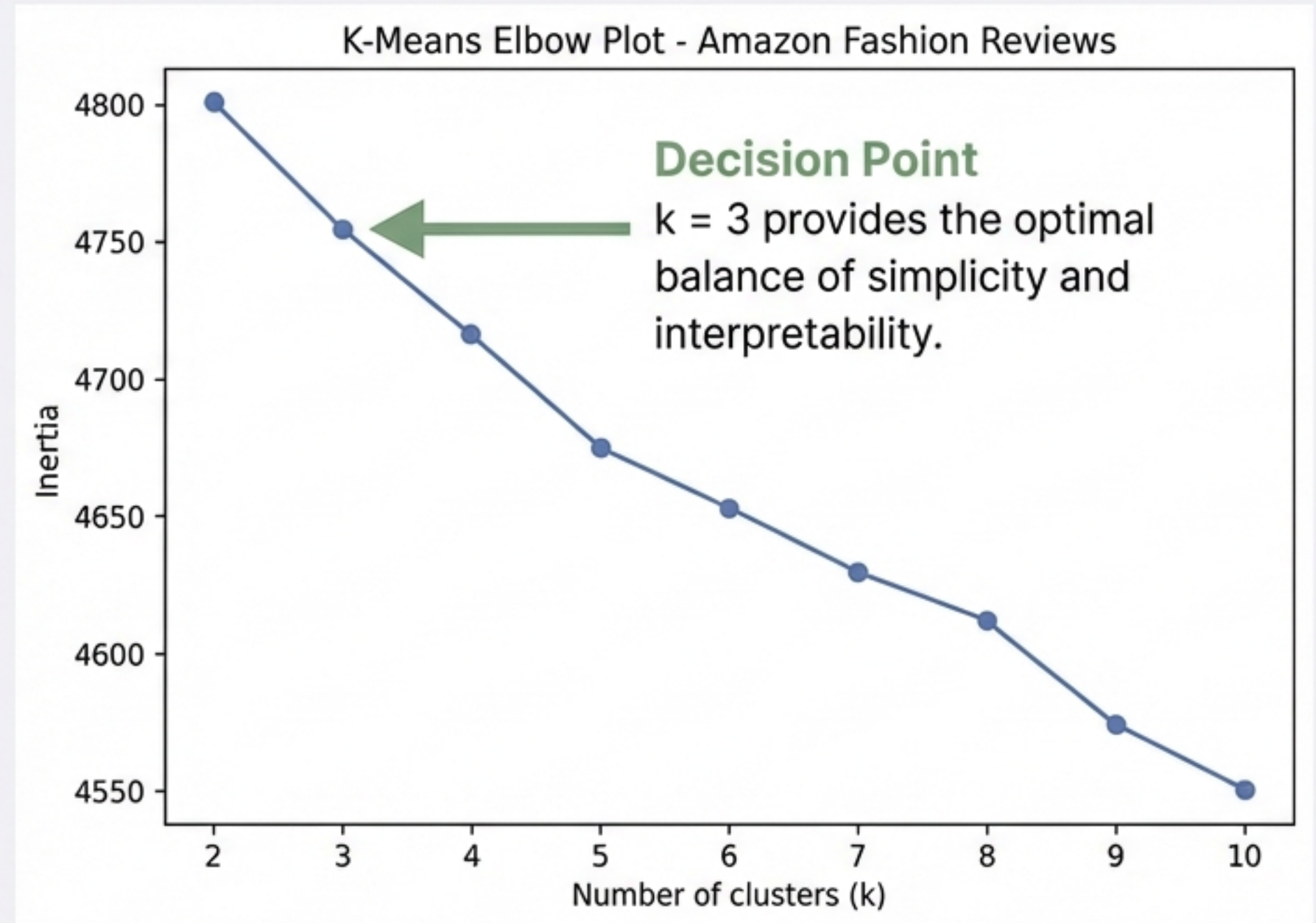
Goal: Group reviews into natural themes without prior labeling.

Technique: K-Means

Partitions the dataset into k distinct, non-overlapping subgroups.

Optimization: The Elbow Method

Used to determine the optimal number of groups by measuring inertia (sum of squared distances).



The Three Dominant Fashion Themes

1. Comfort & Fit

Discussions on sizing, wearability, and physical feel.



2. Quality & Price

Commentary on material value relative to cost.

3. Style & Appearance

Feedback on aesthetics, look, and design.

Validation: Results cross-verified using Agglomerative Hierarchical Clustering.

Phase III Methodology: Sentiment Classification

Goal: Train models to predict Positive (4-5 stars) vs. Negative (1-2 stars) sentiment.

Contenders

Naive Bayes (Probabilistic Baseline)

Support Vector Machine (Linear Classifier)

Decision Tree (Tree-based)

Random Forest (Ensemble Method)

AdaBoost (Boosting Algorithm)

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score

The Winner: Support Vector Machine (SVM)

92.3%
Accuracy

Why it Won

Balanced performance across both positive and negative classes.

Key Metric

Weighted F1-Score: **0.92**

Strength

Effectively handles sparse, high-dimensional data created by text vectorization.

Takeaway: SVM is the most robust choice for this specific text classification task.

The Runner-Up: Random Forest

91.2% Accuracy

****Ensemble Power****

Aggregating multiple decision trees significantly improved results over single trees.

****Performance Profile****

- Strong recall for the positive class.
- Effectively balances bias and variance.

****Verdict****

Only slightly less accurate than SVM, making it a strong alternative.

The Challenge of Class Imbalance: AdaBoost

85.3%
Accuracy (Lowest of Group)

****The Core Issue****

Struggled significantly with the **Negative** class (Low Recall). While it identified positive reviews well, it failed to learn minority class patterns effectively.

****Lesson Learned****

Boosting algorithms can be sensitive to skewed class distributions and noisy text features compared to SVM.

Phase IV: Deep Learning Experiment (LSTM)

****The Concept****

Long Short-Term Memory (LSTM) networks capture the *sequence* of words, understanding context better than bag-of-words models.

****Experimental Setup****

- Sample: 60,000 reviews
- Embedding: 64-dimensional
- Sequence: 100 tokens

Layer (type)	Output Shape	Param #
embedding_2 (Embedding)	(None, 100, 64)	1,280,000
lstm_2 (LSTM)	(None, 64)	33,024
dropout_4 (Dropout)	(None, 64)	0
dense_4 (Dense)	(None, 32)	2,080
dense_5 (Dense)	(None, 1)	0
dropout_4 (Dropout)	(None, 32)	2,080
dropout_5 (Dropout)	(None, 32)	0
dense_5 (Dense)	(None, 1)	33

Total params: 1,315,137 (5.02 MB)
Trainable params: 1,315,137 (5.02 MB)
Non-trainable params: 0 (0.00 B)

LSTM Results and Analysis

81.8%

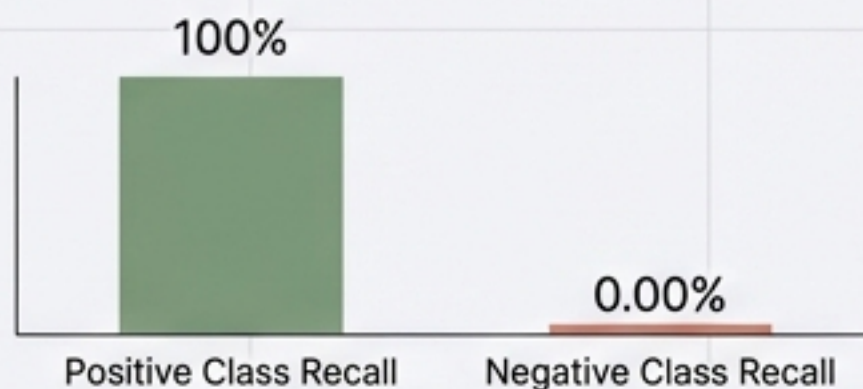
Accuracy

****The Verdict****

Did *not* outperform the traditional SVM model (92.3%).

****Identified Bias****

The model showed **0.00 recall** for Negative reviews in this sample.



****Potential****

Offers promise for capturing nuance, but requires significantly more balanced data and pre-trained embeddings.

Business Implications

****Similarity Analysis****

Auto-detect duplicate reviews and summarize redundant feedback to reduce customer noise.

****Clustering****

Spot product defects immediately.

(e.g., A sudden drop in sentiment within the 'Fit' cluster signals a manufacturing sizing error).

****Classification****

Real-time monitoring.

Models like SVM can flag negative feedback instantly for customer service intervention.

Conclusion: Unlocking the Value of Text

Summary of Success

We successfully transformed **170,000** unstructured reviews into a structured analytical framework.

- **Similarity** identified redundancy and shared patterns.
- **Clustering** revealed the 'Big 3' customer priorities: Fit, Quality, and Style.
- **SVM Classification** provided 92% accuracy in automating sentiment detection.

Text mining offers a scalable, data-driven path to improved decision-making in e-commerce.